1.2.2 Lab Task 2

We constructed the practical differentiator circuit shown in Figure 5 using an LM741 op-amp and breadboard. We configured the power supply as described in the first experiment, and we set the signal generator to produce a triangular wave with a frequency of 2 kHz and an amplitude of 1 V as shown in Figure 16.

Then, to confirm the differentiator circuit's functionality, we measured the voltage output at a chosen time while watching both the input and output waveforms on the oscilloscope screen. We repeated the experiment with square and sinusoidal waves of 1 kHz and 10 kHz frequencies.

1. By analyzing differentiator circuits output on the oscilloscope screen, we found out the given triangular wave form transformed into the square-like waves. Which shows the differentiator circuit works reversed in terms of integrator circuit.

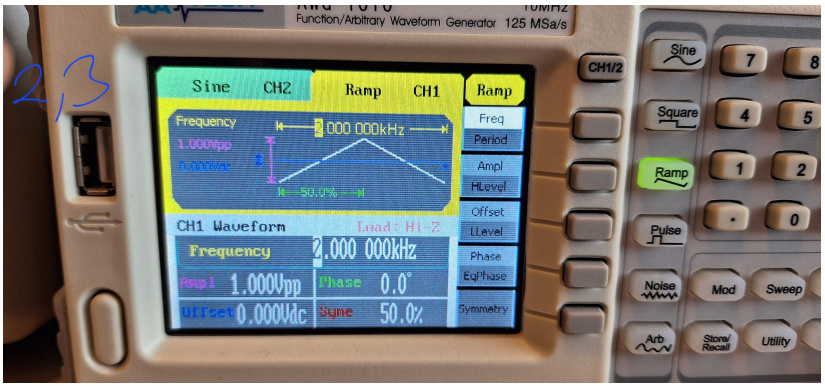


Figure 16: Wave generator generates triangular wave with 2kHz with 1V amplitude.

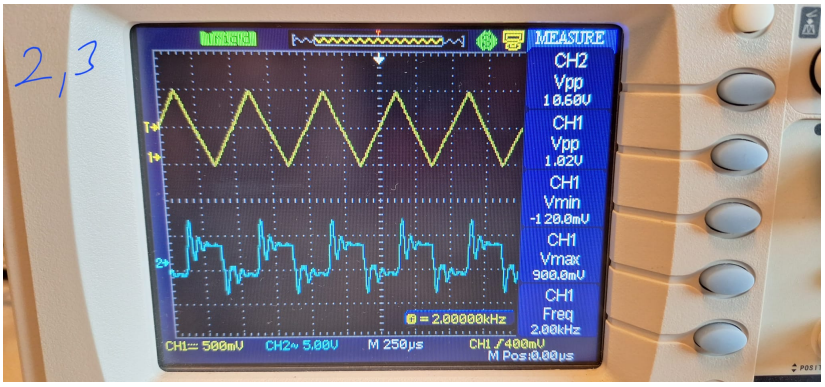


Figure 17: oscilloscope with given triangular input and as output square wave.

2. When we change the input waveform to square wave (Figure 18) and repeat the previous step, with same values.

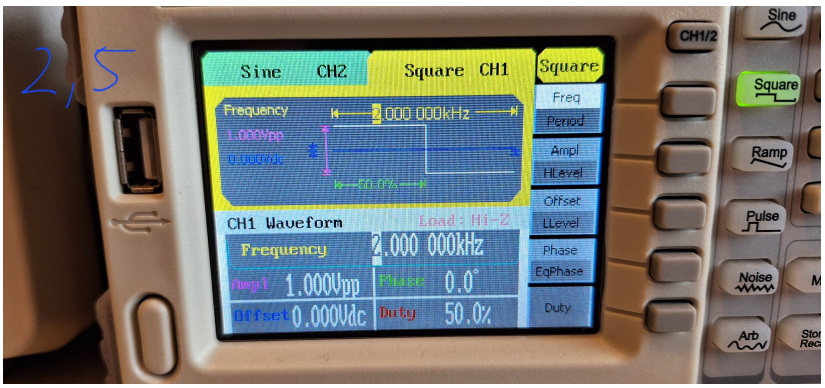


Figure 18: Wave generator generates square wave with 2kHz with 1V amplitude.

3. Using the oscilloscope, we observed both the input and output wave forms on the screen. When we provide square waves we obtain triangular waves as shown in Figure 19

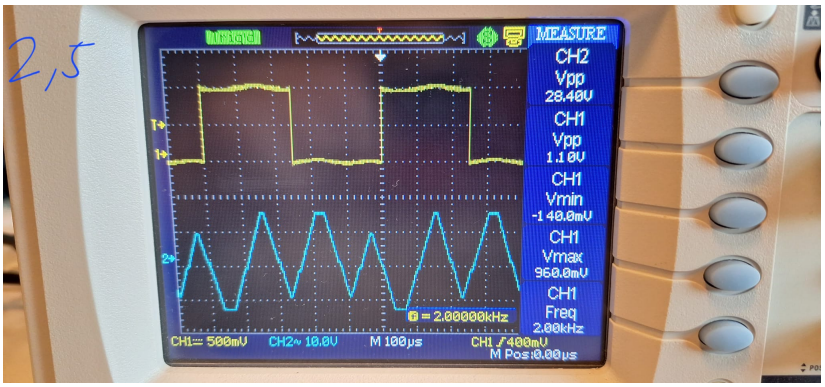


Figure 19: oscilloscope with given square input and as output triangular wave.

4. When we change the input signal to 1 KHz sinusoidal wave (Figure 20) and repeat 1st step.

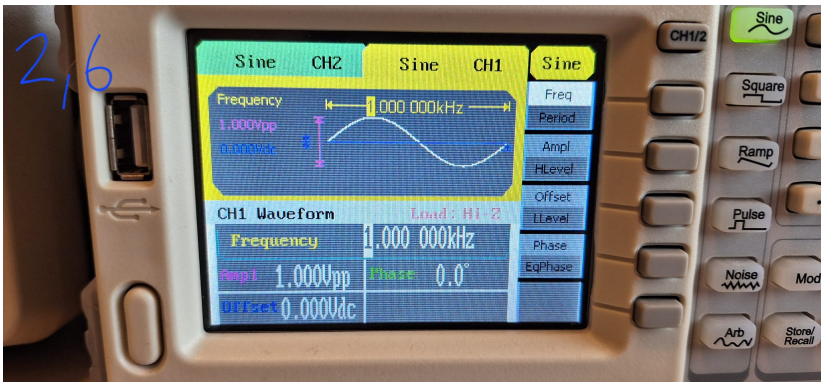


Figure 20: Wave generator generates sinusoidal wave with 1kHz with 1V amplitude.

5. Using the oscilloscope, we observed both the input and output wave forms on the screen. When we provide sinusoidal waves we obtain cosinusoidal waves as shown in Figure 21

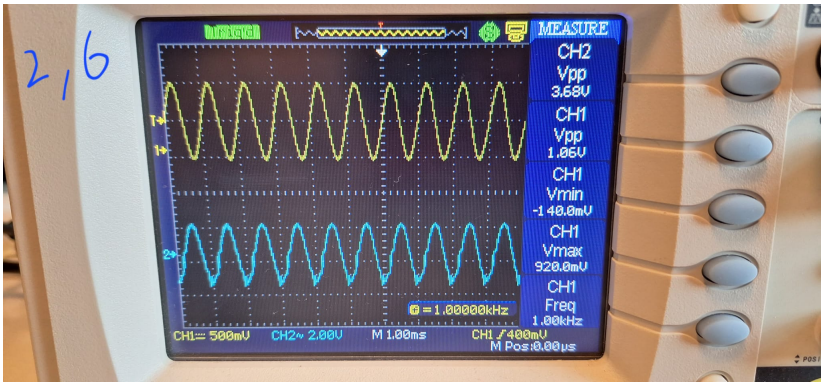


Figure 21: oscilloscope with given sinusoidal input and as output cosinusoidal wave.

6. When we change the input signal to 10 KHz sinusoidal wave (Figure 22) and repeat first step.

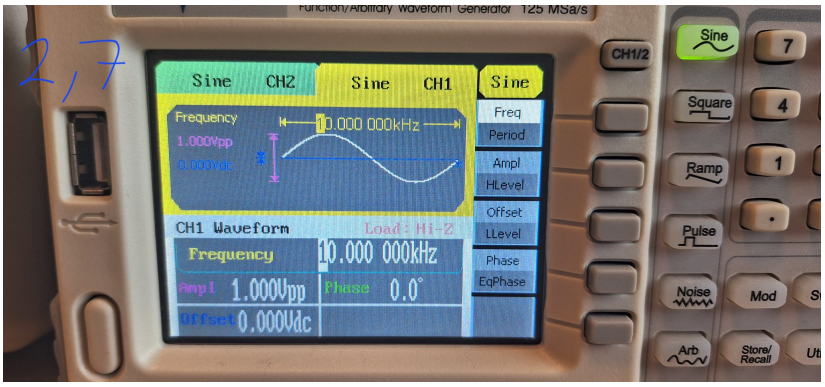


Figure 22: oscilloscope with given sinusoidal input and as output cosinusoidal wave.

7. Using the oscilloscope, we observed both the input and output wave forms on the screen. When we provide sinusoidal waves we obtain triangular waves as shown in Figure 21. As we increase the frequency we obtain graph look like triangular wave.

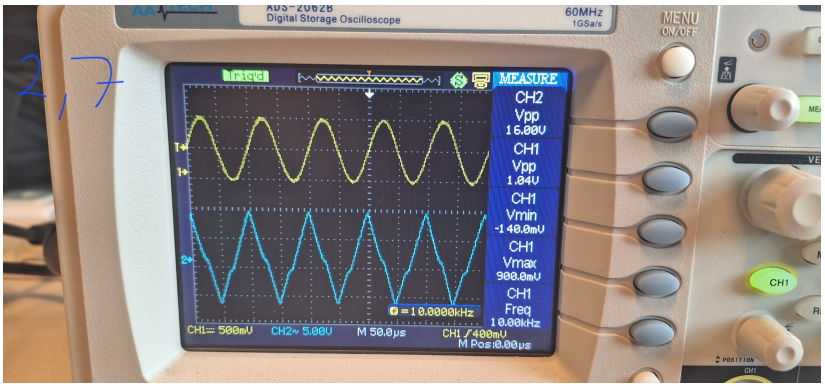


Figure 23: oscilloscope with given sinusoidal input and as output triangular wave.

2 Conclusion

In conclusion we found that the integrator circuit, which included an extra resistor, accurately calculated the integral of the input voltage for different waveforms. Similarly, the practical differentiator circuit, with a resistor in series, effectively reduced noise and accurately represented the derivative of the input voltage. Overall, our experiments improved our understanding of circuit analysis and have practical applications in electronics.